

Skill Specification: Automated Vehicle Analysis

1. Overview

This skill automates "**Vehicle Identification and Analysis**" from recorded video files (MP4) captured by fixed-position cameras. It transforms raw video footage into structured intelligence by extracting vehicle features (make, model, color) and performing OCR on license plates using a combination of computer vision and Multimodal Large Language Models (MLLMs).

2. Objectives

- **Video Processing:** Systematically extract frames from local MP4 files at defined intervals.
- **High-Precision Detection:** Isolate individual vehicles from complex backgrounds using YOLO11.
- **Detailed Attribute Reasoning:** Utilize Qwen2.5-VL to identify vehicle specifics and read license plate information (region and characters).

3. System Architecture

Stage	Tool	Description
Ingestion	OpenCV (cv2)	Reads local MP4 files and samples frames every few seconds to optimize processing load.
Detection	YOLO11n	Scans frames for Class 2 (car) or Class 7 (truck) and generates

Stage	Tool	Description
		cropped images of detected vehicles.
Reasoning	Qwen2.5-VL-7B	Performs visual reasoning on cropped images to extract make, model, color, and license plate details.

4. Key Features

- **Local File Integration:** Optimized for batch processing of stored security or traffic footage without requiring network streaming.
- **State-of-the-Art Detection:** Leverages YOLO11 for faster and more accurate localization of "Involved Vehicles."
- **Contextual OCR:** Uses MLLM capabilities to read license plates even in varying lighting conditions or angles where traditional OCR might fail.
- **Privacy-Conscious Analysis:** Uses neutral terminology for vehicle identification to ensure objective reporting.

5. Requirements

- **Hardware:** NVIDIA GPU with 16GB+ VRAM (e.g., Tesla T4, A100, or local RTX series).
- **Libraries:** ultralytics (YOLO11), transformers, qwen-vl-utils, opencv-python, torch.

6. Usage

1. **Set Input Path:** Place your MP4 video files in the designated source folder.

2. **Execute Pipeline:** Run the processing script to trigger frame extraction and vehicle cropping.
3. **Review Intelligence:** Access the generated reports containing cropped visual evidence alongside AI-extracted metadata.